

PGA-UNet: A Lightweight Prompt-Guided Attention U-Net for Interactive Bone X-Ray Lesion Segmentation

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ABSTRACT Bone X-ray lesion segmentation remains difficult because lesions are often small, low-contrast, and anatomically confounded, so fully automatic segmentation models may fail before accurate boundary prediction becomes possible. Prior studies addressed this problem either with automatic encoder-decoder networks such as U-Net and Attention U-Net or with promptable foundation models such as SAM-Med2D. However, automatic models do not exploit user-provided spatial guidance, whereas foundation models are computationally heavy and do not necessarily preserve prompt influence throughout a lightweight task-specific network. This article proposes PGA-UNet, a lightweight prompt-guided convolutional architecture that converts a bounding-box prompt into a Gaussian-smoothed plateau heatmap, amplifies prompt-relevant encoder features through a Prompt Spatial Gate (PSG), and conditions decoder skip attention through a Conditional Attention Decoder (CAD). On BTXRD and FracAtlas, PGA-UNet achieved Dice scores of 0.8788 and 0.8286 at 512×512 under covering prompts, respectively. In a matched 256×256 comparison, PGA-UNet outperformed fine-tuned SAM-Med2D under both covering and off-center prompts on both datasets, despite being trained only on covering prompts while SAM-Med2D was trained on both, with the clearest advantage on the 50 smallest BTXRD lesions, where fine-tuned SAM-Med2D collapsed to a Dice of 0.1903 while PGA-UNet reached 0.7650 at the same resolution. Ablation results further showed that prompt-gated encoder features (PSG) and prompt-conditioned decoder attention (CAD) are most effective when combined rather than in isolation, and that the full architecture also makes the Gaussian-smoothed prompt more beneficial than a binary prompt. PGA-UNet uses 2.95M parameters, corresponding to an approximately 92-fold reduction in parameter count, an 11.9-fold reduction in FLOPs, and a 6.0-fold reduction in GPU inference latency relative to SAM-Med2D in the tested 256×256 setting. These results indicate that prompt information is most effective in this task when it is both spatially softened and propagated throughout the network from early encoding to late decoding.

INDEX TERMS bone X-ray, interactive segmentation, medical image segmentation, prompt-guided learning, Attention U-Net

I. INTRODUCTION

BONE X-ray imaging is widely used in screening, diagnosis support, and follow-up because it is fast, inexpensive, and broadly available. However, lesion segmentation in radiographs remains difficult because targets may be small, low-contrast, elongated, or partially occluded by surrounding anatomy. In such cases, the main failure often occurs at localization: if the model attends to the wrong anatomical region, accurate boundary prediction becomes irrelevant.

This difficulty motivated two broad development lines in prior work. One line improved fully automatic segmentation

networks, beginning with U-Net-style encoder-decoder architectures and later extending them with attention mechanisms such as Attention U-Net [3], [4]. These models remain strong and efficient baselines, but they still solve a harder problem than many practical workflows require, namely localizing and segmenting every lesion from scratch. A second line moved toward interactive or prompt-guided segmentation, where clicks, scribbles, geodesic cues, or boxes are provided by the user [5], [6]. More recently, promptable foundation models such as SAM and SAM-Med2D pushed this direction further [1], [2]. Although powerful, these models are heavy

and may be unnecessarily expensive for a fixed grayscale radiograph task with repeated low-latency interaction.

In the present setting, a clinician can often provide a coarse bounding box more easily than a fully automatic system can localize a small lesion reliably. This creates a more focused research question: how should a lightweight model use that coarse spatial cue? A simple answer is to concatenate the prompt as an extra channel. However, that strategy does not guarantee that the prompt remains influential once features move deeper through the network, especially when the prompt is slightly misaligned or when lesion evidence is weak.

This article therefore investigates a lightweight alternative. We propose PGA-UNet, a prompt-guided convolutional architecture that treats the box prompt as a structured spatial prior rather than as a simple auxiliary channel. The prompt is encoded as a plateau-style Gaussian heatmap, injected into encoder features through a Prompt Spatial Gate (PSG), and reused in the decoder through a Conditional Attention Decoder (CAD). The underlying encoder-decoder structure remains U-Net-like, but prompt information is not limited to the input layer; it is amplified and maintained throughout the network.

The main contributions of this article are as follows.

- We present a compact prompt-guided architecture for interactive bone X-ray lesion segmentation.
- We integrate plateau-style Gaussian prompting, encoder-side prompt gating, and decoder-side prompt-conditioned attention within a lightweight network.
- We formulate an offline training and online interactive inference framework for prompt-guided radiographic lesion segmentation, with explicit mathematical definitions for prompt encoding, encoder gating, and decoder conditioning.
- We evaluate the method on BTXRD and FracAtlas under covering and off-center prompt settings, including comparisons with Attention U-Net and SAM-Med2D, a small-lesion analysis, Monte Carlo cross-validation, and efficiency measurements.
- We report ablation, efficiency, and small-lesion results that clarify where the proposed prompt-conditioning design is most useful.

II. RELATED WORK

Automatic medical image segmentation progressed from fully convolutional encoder-decoder architectures toward more selective skip-feature processing. U-Net established the canonical multi-scale encoder-decoder template for biomedical segmentation [3]. Attention U-Net then introduced attention gates on skip connections so that decoder-side fusion could suppress irrelevant spatial responses more selectively [4]. This line of work improved automatic segmentation quality, but it still assumes that the network itself must both localize and segment the lesion from the raw image alone.

Interactive medical segmentation developed in parallel from the observation that partial user guidance can simplify

TABLE 1. Representative prior work and the remaining gap addressed in this article.

Work	Principle	Method	Performance Evidence	Pros	Cons
U-Net [3]	Fully convolutional encoder-decoder	Multi-scale skip-connected segmentation network	Broad biomedical benchmarks: Dice-based overlap reporting	Simple, efficient, strong canonical baseline	No prompt guidance; all localization must be automatic
Attention U-Net [4]	Skip-feature filtering by attention gates	U-Net with gated skip connections	Medical image segmentation tasks with overlap metrics such as Dice and IoU	Stronger automatic baseline than plain U-Net	Attention is still learned without explicit prompt conditioning
DeepGeoS and related interactive methods [6]	User guidance reduces search space	Geodesic or interaction-aware segmentation	Interactive medical segmentation under user input	Better reflects clinical interaction	Interaction mechanism can be task-specific or expensive to maintain
SAM-Med2D / MedSAM [2], [9]	Promptable foundation segmentation	Large pretrained model with prompt-driven mask decoding	Multi-dataset medical evaluation with prompt-based metrics	Flexible and broadly applicable	Heavy; prompt influence is not designed as a lightweight end-to-end radiograph-specific prior
This work	Prompt as a dense spatial prior throughout the network	Gaussian plateau prompt + PSG + CAD in a lightweight CNN	BTXRD and FracAtlas: Dice, IoU, HD95, CBL, efficiency, small-lesion analysis	Lightweight, prompt-aware from encoder to decoder, robust to prompt perturbation	Still requires supervised masks and fixed-resolution preprocessing

the problem substantially [5], [6]. Early methods used clicks, scribbles, and geodesic cues; later work incorporated boxes as practical coarse prompts. The next major step was promptable foundation models such as SAM, SAM-Med2D, and MedSAM, which generalized prompt-driven segmentation across many medical domains [1], [2], [9]. These models are flexible, but their prompt usage is architecturally different from lightweight task-specific CNNs, and their computational cost is much higher. For very small radiographic lesions, another unresolved issue is whether a coarse prompt merely indicates the approximate region or whether it continues to shape feature extraction and decoding throughout the network.

The present work is motivated by that gap. In grayscale bone radiographs, the challenge is not only to know *where* the lesion roughly is, but also to preserve weak lesion evidence while suppressing surrounding clutter. A lightweight prompt-guided CNN may therefore benefit from making prompt information active from early encoding to late decoding instead of using the prompt only as an initial cue.

III. METHOD

A. OFFLINE AND ONLINE FRAMEWORK

The proposed system has two stages. In the offline stage, the model is trained on radiographs paired with lesion polygons. Each polygon is converted into a training prompt by generating a simulated bounding box and then transforming that box into a Gaussian-smoothed plateau heatmap. The network learns to map the image-prompt pair to the binary lesion mask. In the online stage, a user provides a coarse bounding box around a suspected lesion on a new radiograph. The same prompt-construction procedure is applied, and the trained model predicts the final lesion mask.

The scientific contribution of this framework is that the prompt is not treated as an optional side input used only at the first layer. Instead, the prompt is encoded as a dense spatial prior and made active at both encoder and decoder stages. The practical consequence is improved robustness when the prompt is coarse, mildly off-center, or applied to a very small lesion.

B. PROBLEM SETTING

Given a grayscale radiograph $\mathbf{I} \in \mathbb{R}^{H \times W}$ and a user-provided bounding box $\mathbf{B}$, the goal is to predict a binary lesion mask $\hat{\mathbf{M}}$. Let $\mathbf{H}(\mathbf{B})$ be the prompt heatmap derived from the box. PGA-UNet predicts

$$\hat{\mathbf{M}} = 1 [f_{\text{PGA}}(\mathbf{I}, \mathbf{H}(\mathbf{B}); \theta) \geq 0.5]. \quad (1)$$

C. PROMPT REPRESENTATION

The input prompt is not encoded as a hard binary rectangle. Instead, the box interior is converted into a plateau map and its boundaries are softened with Gaussian filtering. In the implementation used for the reported experiments, the box is first assigned a minimum prompt margin before rasterization and then smoothed with a Gaussian kernel. At 256×256 , the minimum margin is 5 pixels and the kernel size is 31×31 ; at 512×512 , the corresponding values are 10 pixels and 61×61 . This preserves the full spatial extent of the region of interest while reducing over-dependence on an abrupt boundary.

During evaluation, prompts are tested in two settings. A covering prompt expands the lesion bounding box by 15%–45% of lesion width and height. An off-center prompt first applies the same expansion and then shifts the box by up to 30% of lesion width and height while still covering the lesion. A single model is trained per dataset and resolution; the prompt condition is varied only at evaluation time unless otherwise stated in the ablation study.

Let (x_1, y_1, x_2, y_2) denote the tight lesion box and let $w = x_2 - x_1$ and $h = y_2 - y_1$. The simulated covering box expands each side by a random fraction in $[0.15, 0.45]$ of w or h during training and by a fixed fraction of 0.30 during evaluation. The off-center box applies the same expansion and then adds a translation bounded by $0.30w$ horizontally and $0.30h$ vertically while preserving lesion coverage. This simulation is not a substitute for radiologist-drawn prompts, but it provides a controlled way to test sensitivity to plausible box variation.

Algorithmically, prompt construction follows four steps: 1) derive a lesion box from the polygon annotation or user input, 2) enlarge that box with a resolution-aware minimum margin, 3) rasterize the enlarged region into a plateau mask, and 4) smooth the plateau boundary with a Gaussian kernel. This sequence avoids the hard-edge dependence of a binary box while preserving a clear spatial prior.

D. PROMPT SPATIAL GATE

Let $\mathbf{x}^l$ denote the encoder feature map at level l . PSG modulates it with a prompt-derived attention map $\mathbf{A}_{\text{PSG}}^l$:

$$\tilde{\mathbf{x}}^l = \mathbf{x}^l \odot (1 + \alpha_{\text{PSG}}^l \mathbf{A}_{\text{PSG}}^l), \quad (2)$$

where $\odot$ is element-wise multiplication and α_{PSG}^l is learnable. In the implementation, α_{PSG}^l is initialized to 0.1 and clamped to $[0, 1]$. The residual form keeps the original feature stream available while emphasizing prompt-relevant regions.

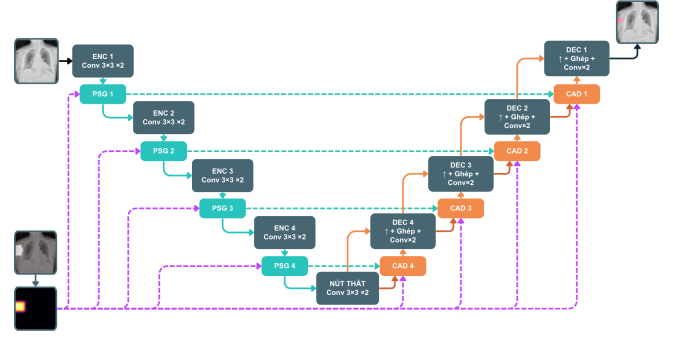


FIGURE 1. Overview of PGA-UNet. Prompt information derived from the bounding box is injected into encoder features through PSG and reused in decoder skip attention through CAD.

E. CONDITIONAL ATTENTION DECODER

CAD extends decoder attention by conditioning the skip gate on prompt-encoded features. Let $\mathbf{g}^l$ be the decoder gating feature and $\mathbf{p}_{\text{enc}}^l$ the prompt encoding at the same scale. The conditioned gate is

$$\mathbf{g}'^l = \mathbf{g}^l + c^l \alpha_{\text{CAD}}^l w_l \mathbf{p}_{\text{enc}}^l, \quad (3)$$

where c^l is a prompt confidence scalar, α_{CAD}^l is learnable, and w_l is a fixed depth coefficient. This design helps the decoder remain aligned with the prompt even when the box is not perfectly centered.

The prompt encoder in CAD uses two 3×3 convolutions with instance normalization and ReLU. The prompt-confidence scalar c^l is produced by global average pooling followed by a 1×1 convolution and a sigmoid nonlinearity. The decoder depth coefficients are fixed to $\{1.0, 0.7, 0.4, 0.2\}$ from deep to shallow layers. With feature scale 4, the channel widths are $\{16, 32, 64, 128, 256\}$ from the shallowest encoder stage to the bottleneck. The CAD mixing coefficient is parameterized as a sigmoid-transformed scalar and initialized near 0.3.

Taken together, the method can be summarized as follows: 1) preprocess the image and prompt into a common square resolution, 2) encode prompt-relevant spatial priors with PSG in the encoder, 3) propagate prompt-conditioned context through CAD in the decoder, and 4) predict the final binary lesion mask. The scientific meaning of the contribution is that prompt information is both spatially softened and repeatedly reused, rather than being injected once and left to dissipate.

IV. EXPERIMENTAL SETUP

A. DATASETS

We evaluate on two public bone radiograph datasets. BTXRD is a primary bone tumor X-ray dataset with polygon annotations for classification, localization, and segmentation [7]. Our BTXRD split contains 1,493/187/187 images with 1,848/238/232 lesion polygons for training, validation, and testing, respectively. FracAtlas is a fracture radiograph

dataset with classification, localization, and segmentation labels [8]. After data cleaning and polygon normalization, our FracAtlas split contains 573/72/72 images with 730/95/92 lesion polygons for training, validation, and testing, respectively.

Splits are performed at the image level, and all polygons from the same image are kept in the same split. The public metadata were insufficient to verify strict patient-level separation. Each polygon is treated as one promptable lesion sample during training. For image-level evaluation, if an image contains multiple lesions, the model is run once per lesion box and the probability maps are merged by pixelwise maximum before thresholding at 0.5. Ground-truth masks are merged by union in the same image.

Inputs are resized isotropically and padded to square resolutions of 256×256 or 512×512 to preserve aspect ratio. Images, masks, and prompt maps undergo the same geometric preprocessing.

B. TRAINING DETAILS

All CNN models were implemented in PyTorch and trained on an NVIDIA Tesla T4 GPU with 16 GB memory. PGA-UNet and Attention U-Net were trained separately on each dataset and resolution. The optimizer was AdamW with learning rate 10^{-4} and weight decay 10^{-4} . The loss was the sum of binary cross-entropy and Dice loss. Batch size was 4, the maximum training budget was 100 epochs, and early stopping with patience 15 was used. The primary model-selection criterion was validation Dice under the covering-prompt condition.

Training-time augmentation included horizontal flipping and random rotation within $\pm 15^\circ$. The implementation also applied prompt dropout or prompt noise to a subset of training iterations to reduce over-reliance on an ideal prompt. PGA-UNet itself is trained only under the covering-prompt (zoom-out) condition; the off-center condition is introduced only at evaluation time, so off-center testing measures generalization to a prompt perturbation the network never saw during training. For the fine-tuned SAM-Med2D comparison, the pre-trained image encoder was frozen except for its lightweight Adapter layers, while the prompt encoder and mask decoder were fully updated on each dataset under the same split protocol. Rather than the original authors' training-time box perturbation, a tight box with a few pixels of random jitter combined with iterative point-based refinement, SAM-Med2D was finetuned box-only, with each training box independently drawn on a per-sample 50/50 basis from the same covering-prompt protocol used for PGA-UNet: a random expansion of 15%–45% per side, or the same expansion followed by an off-center shift of up to 30%. Unlike PGA-UNet, SAM-Med2D is therefore trained on both the covering and off-center conditions, so its off-center test performance reflects an in-distribution condition rather than the out-of-distribution generalization being tested for PGA-UNet. This asymmetry is intentional: each baseline is trained the way it is normally trained or, in SAM-Med2D's case, trained under a proto-

col matched to PGA-UNet's covering-prompt distribution to avoid confounding architectural differences with an unrelated train-test distribution shift, while both models are evaluated under the same covering and off-center test protocol for a fair head-to-head comparison. Validation during SAM-Med2D finetuning and the final test split used the same covering-prompt protocol as PGA-UNet, with batch size 4, a maximum of 100 epochs, early stopping with patience 30, and learning rate 10^{-5} . Additional stability experiments were performed using Monte Carlo cross-validation, that is, repeated random image-level splits rather than strict non-overlapping k -fold partitioning.

Two evaluation details are important for interpreting the results. First, prompt generation is resolution-aware: at 256×256 the minimum prompt margin is 5 pixels with a Gaussian kernel size of 31, while at 512×512 the corresponding values are 10 pixels and 61. Second, the Monte Carlo cross-validation experiments are auxiliary stability evidence rather than replacements for the fixed train/validation/test split used in the main tables.

C. BASELINES AND METRICS

PGA-UNet is compared with Attention U-Net [4] as the main automatic baseline and with SAM-Med2D [2] as the main prompt-based baseline. Attention U-Net is trained without prompts, whereas SAM-Med2D is evaluated at 256×256 in fine-tuned form with the same dataset splits and the same box prompts as PGA-UNet.

We report Dice, IoU, precision, recall, HD95, and center-based localization (CBL). For a predicted mask centroid (x_p, y_p) , a ground-truth centroid (x_g, y_g) , and the diagonal length D_{GT} of the ground-truth bounding box, CBL is defined as

$$\text{CBL} = \max \left(0, 1 - \frac{\sqrt{(x_p - x_g)^2 + (y_p - y_g)^2}}{D_{GT}} \right). \quad (4)$$

HD95 is computed from bidirectional boundary distances. If both masks are empty, HD95 is set to 0. If exactly one mask is empty, HD95 is set to the image side length (256 or 512 pixels). No samples are discarded.

D. SCOPE OF ANALYSIS

This article focuses on prompt-guided segmentation itself and excludes the auxiliary screening pipeline explored separately in the thesis manuscript. To keep the article compact, we retain only the subgroup analyses that most directly support the main claim: robustness under imperfect prompts and effectiveness on small lesions.

V. RESULTS

A. MAIN COMPARISON ACROSS DATASETS AND RESOLUTIONS

This first group of experiments addresses the most basic question: does PGA-UNet remain effective across both datasets, across both tested input resolutions, and across both prompt

TABLE 2. PGA-UNet Dice under covering and off-center prompts at two resolutions.

Dataset	Resolution	Covering	Off-center
BTXRD	256×256	0.8449	0.8111
BTXRD	512×512	0.8788	0.8496
FracAtlas	256×256	0.7974	0.7619
FracAtlas	512×512	0.8286	0.7925

scenarios? These experiments primarily validate the main system-level contribution, namely that prompt-conditioned processing through both encoder and decoder can be transferred across two radiographic datasets with different lesion characteristics. PGA-UNet performs well on both BTXRD and FracAtlas under the intended workflow with coarse box guidance. Across the main experiments, the model remains competitive at both 256×256 and 512×512 and under both covering and off-center prompt settings. The consistent trend across datasets, resolutions, and prompt scenarios is that explicit prompt conditioning in both the encoder and decoder remains useful even when lesion appearance changes substantially between the two datasets.

B. EFFECT OF INPUT RESOLUTION

This experiment isolates the contribution of image resolution within the same architecture and prompt protocol. It supports the claim that the proposed method is especially useful when small lesion evidence would otherwise be weakened by down-sampling. PGA-UNet was trained independently at 256×256 and 512×512 on each dataset. The higher resolution consistently improved Dice under both prompt conditions, and the gain was slightly larger on BTXRD than on FracAtlas. This difference is consistent with the datasets themselves: BTXRD contains many small or irregular lesions whose local evidence degrades noticeably after aggressive downsampling, whereas FracAtlas more often contains elongated but visually simpler fracture structures.

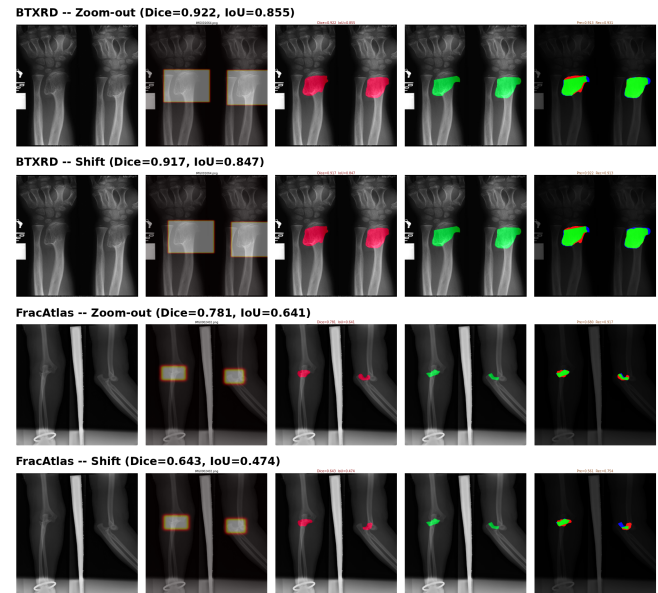
The resolution comparison should not be overinterpreted as a pure image-size effect because the prompt representation also changes with resolution through its margin and Gaussian smoothing scale. Even with that caveat, the main trend remains useful: prompt-guided segmentation of small radiographic lesions benefits from preserving more spatial detail, especially when the lesion occupies only a tiny fraction of the image.

C. MONTE CARLO CROSS-VALIDATION

This experiment examines whether the main findings remain stable when the image-level split is redrawn repeatedly under limited data. It supports the robustness claim at the dataset-partition level rather than at the prompt-perturbation level. Supplementary experiments therefore used Monte Carlo cross-validation, that is, four repeated random image-level splits of the 512×512 PGA-UNet configuration, evaluated

TABLE 3. Monte Carlo cross-validation for PGA-UNet at 512×512 (4 repeated random splits, mean $\pm$ standard deviation).

Dataset	Prompt	Dice	IoU	HD95
BTXRD	Covering	0.8715 ± 0.0083	0.7799 ± 0.0117	8.09 ± 0.67
	Off-center	0.8357 ± 0.0120	0.7260 ± 0.0164	10.45 ± 0.65
FracAtlas	Covering	0.8281 ± 0.0052	0.7140 ± 0.0077	6.93 ± 0.08
	Off-center	0.7862 ± 0.0114	0.6561 ± 0.0154	8.75 ± 0.48

**FIGURE 2.** Qualitative behavior under prompt perturbation. PGA-UNet remains more stable when the prompt is shifted away from the ideal covering box, although thin fracture patterns are still harder than larger compact lesions.

separately under covering and off-center prompts. These results are not intended to replace the fixed-split main tables, but they provide additional evidence that PGA-UNet remains competitive across repeated random repartitions rather than depending on a favorable single split.

The standard deviation across the four repeated splits stays under 0.012 Dice in every row, and the covering-versus-off-center ordering established in the main split (Table 2) is preserved in every individual fold. This indicates that the resolution and prompt-condition trends reported above are not an artifact of a single favorable train/validation/test partition.

D. COMPARISON WITH THE AUTOMATIC BASELINE

This comparison tests whether the proposed prompt-guided architecture yields a measurable advantage over a strong automatic encoder-decoder baseline. The contribution being examined here is not merely model size, but the use of coarse prompt information throughout the network. Table 4 compares PGA-UNet with Attention U-Net at 512×512 . This comparison is intentionally between a prompt-guided model and a strong automatic baseline without prompt guidance. PGA-UNet achieves higher system-level segmentation performance when coarse box guidance is available on both BTXRD and FracAtlas.

TABLE 4. Comparison with the automatic baseline at 512×512 under covering prompts.

Dataset	Model	Dice	IoU	HD95
BTXRD	Attention U-Net	0.5160	0.4137	129.67
	PGA-UNet	0.8788	0.7895	9.08
FracAtlas	Attention U-Net	0.3075	0.2189	110.83
	PGA-UNet	0.8286	0.7126	6.21

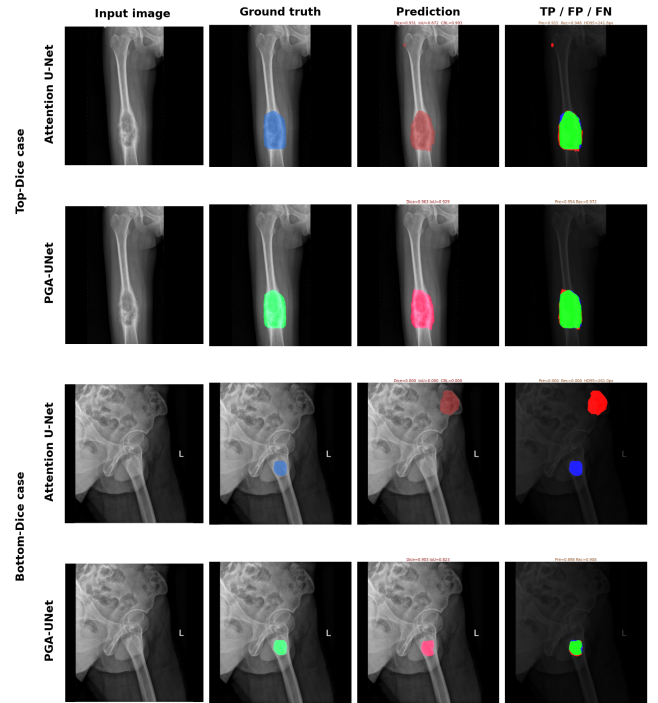
TABLE 5. Top-Dice and bottom-Dice comparison under covering prompts at 512×512 (50 images per group).

Dataset	Group	Model	Dice	HD95
BTXRD	Top-Dice	Attention U-Net	0.8629	39.32
		PGA-UNet	0.9036	9.23
	Bottom-Dice	Attention U-Net	0.0335	270.72
		PGA-UNet	0.8521	5.19
FracAtlas	Top-Dice	Attention U-Net	0.4428	82.08
		PGA-UNet	0.8318	6.37
	Bottom-Dice	Attention U-Net	0.1529	135.76
		PGA-UNet	0.8151	6.25

E. ATTENTION U-NET TOP- AND BOTTOM-DICE SUBSETS

This subgroup analysis is an extension of the automatic-baseline comparison. After comparing PGA-UNet with Attention U-Net at the dataset level, we ask a more specific question: how does PGA-UNet behave on images where the automatic baseline is relatively strong versus relatively weak? The purpose is to clarify the practical meaning of the Attention U-Net comparison rather than to define objective data difficulty. We therefore compare Attention U-Net top-Dice and bottom-Dice image groups on both BTXRD and FracAtlas. Specifically, for each dataset we select the 50 test images with the highest Attention U-Net Dice scores and the 50 test images with the lowest Attention U-Net Dice scores, then re-evaluate PGA-UNet on the same two groups. This analysis does not define intrinsic data difficulty; it only reports performance on subsets induced by the Attention U-Net ranking.

On BTXRD, PGA-UNet improves over Attention U-Net even in the top-Dice group, raising Dice from 0.8629 to 0.9036. The contrast becomes much larger in the bottom-Dice group, where Attention U-Net collapses to 0.0335 Dice while PGA-UNet still reaches 0.8521. FracAtlas shows the same qualitative pattern with a different baseline severity: even the Attention U-Net top-Dice group only reaches 0.4428, well below its BTXRD counterpart, which indicates that the automatic baseline struggles more broadly on FracAtlas; PGA-UNet nonetheless reaches 0.8318 on that same group and remains close to that level, at 0.8151, on the bottom-Dice group. Because the four groups are defined directly from Attention U-Net scores, these values should not be interpreted as global averages; instead, they highlight how PGA-UNet behaves on cases where the automatic baseline is relatively strong or weak.

**FIGURE 3.** Qualitative comparison between Attention U-Net and PGA-UNet on BTXRD for one case from the Attention U-Net top-Dice group and one case from the Attention U-Net bottom-Dice group.

The qualitative examples help clarify this behavior. In the Attention U-Net top-Dice groups, the automatic baseline can often place the predicted mask in roughly the correct anatomical region, but it may still produce spurious responses far from the lesion. Even when the lesion itself is covered reasonably well, those distant false positives reduce overlap-based metrics such as Dice and IoU, which is consistent with the still-elevated HD95 values (39.32 on BTXRD, 82.08 on FracAtlas) despite a moderate-to-low Dice. PGA-UNet is less prone to this pattern because the bounding box prompt constrains the model to a more relevant spatial region. In the bottom-Dice groups, the failure is more severe: the automatic baseline may localize the wrong region or miss the lesion almost entirely. A likely reason is that the attention gate itself has no guarantee of learning a clinically correct gating policy: it filters whatever features it has learned to emphasize, whether those features correspond to lesions or to spurious high-energy structures. When the learned response is dominated by background clutter or artifact-like signals with strength comparable to, or larger than, the true lesion evidence, the gate can no longer rescue localization. In practice, it may suppress only mild background activity while remaining ineffective against stronger false responses. In contrast, PGA-UNet receives a coarse spatial prior and can therefore identify the lesion region more reliably before refining the mask.

TABLE 6. Dice comparison with fine-tuned SAM-Med2D at 256×256 .

Dataset	Prompt	PGA-UNet	SAM-Med2D-FT
BTXRD	Covering	0.8449	0.7624
	Off-center	0.8111	0.7542
FracAtlas	Covering	0.7974	0.7439
	Off-center	0.7619	0.7275

F. MATCHED COMPARISON WITH SAM-MED2D

This comparison tests the architectural contribution of the proposed prompt handling against a prompt-based foundation baseline under matched input conditions. Both models receive the same prompt type and the same resolution, so the difference is more directly tied to how prompt information is represented and propagated. Table 6 reports Dice at 256×256 under both covering and off-center prompts. PGA-UNet achieved higher Dice than fine-tuned SAM-Med2D under both prompt conditions on both datasets. The small-lesion analysis presented next should be read as an extension of this SAM-Med2D comparison to the setting where prompt usage is expected to matter most.

Under covering prompts, PGA-UNet outperforms fine-tuned SAM-Med2D by 0.0825 Dice on BTXRD and 0.0535 Dice on FracAtlas at the same 256×256 resolution; under off-center prompts the margin is 0.0569 and 0.0344 Dice, respectively. PGA-UNet therefore remains ahead in absolute Dice under both prompt conditions on both datasets.

The relative degradation from covering to off-center prompts is worth reading carefully rather than as a simple robustness ranking. PGA-UNet's own drop is 0.0338 Dice on BTXRD and 0.0355 Dice on FracAtlas (Table 2), while fine-tuned SAM-Med2D's drop is smaller, 0.0082 and 0.0164 Dice, respectively. Taken at face value this could suggest that SAM-Med2D degrades less under prompt perturbation, but the two models were not trained on the same prompt distribution: as described in Section IV-B, PGA-UNet is trained only under the covering-prompt condition and is evaluated on off-center prompts it never saw during training, whereas fine-tuned SAM-Med2D is trained on a 50/50 mixture of covering and off-center prompts, so the off-center test condition is in-distribution for SAM-Med2D but out-of-distribution for PGA-UNet. Despite this asymmetry working against it, PGA-UNet still achieves higher absolute Dice than fine-tuned SAM-Med2D under off-center prompts on both datasets. This is consistent with the view that PGA-UNet's off-center behavior comes from how PSG and CAD propagate and reuse prompt information architecturally, rather than from having been trained on the perturbation being tested.

G. PERFORMANCE ON SMALL LESIONS

This experiment is an extension of the SAM-Med2D comparison to the setting in which the contribution of prompt-aware feature preservation should be most visible, namely very small lesions. To keep subgroup analysis objective, we retain

TABLE 7. Dice and HD95 on the 50 smallest lesions per dataset.

Dataset	Model	Dice	HD95
BTXRD	SAM-Med2D-FT 256×256	0.1903	240.72
	PGA-UNet 256×256	0.7650	3.11
	PGA-UNet 512×512	0.8483	2.32
FracAtlas	SAM-Med2D-FT 256×256	0.3736	105.19
	PGA-UNet 256×256	0.7973	5.62
	PGA-UNet 512×512	0.8348	4.83

only the small-lesion subset because it is defined directly by lesion size rather than by a model-dependent ranking rule. For each dataset, the subset is formed by selecting the 50 test images with the smallest total lesion area, evaluated under covering prompts. This subset is then used for two related comparisons: PGA-UNet versus fine-tuned SAM-Med2D at the matched resolution of 256×256 , and PGA-UNet₂₅₆ versus PGA-UNet₅₁₂ to examine the effect of resolution. All metrics are computed at a shared 512×512 reference frame, with the 256×256 predictions of SAM-Med2D and PGA-UNet upsampled before scoring, so that the three models are compared on identical ground-truth masks.

On the BTXRD small-lesion subset, PGA-UNet at 256×256 improves Dice by 0.5747 over fine-tuned SAM-Med2D at the same resolution, and increasing the PGA-UNet input size to 512×512 yields a further gain of 0.0833 Dice. The HD95 gap is even more striking: fine-tuned SAM-Med2D averages 240.72 pixels, close to the maximum possible penalty on a 512×512 frame, indicating that it frequently misses the small lesion entirely rather than producing a merely imprecise boundary. On the FracAtlas small-lesion subset the same pattern holds at a smaller magnitude: PGA-UNet at 256×256 improves Dice by 0.4237 over fine-tuned SAM-Med2D, with a further 0.0375 Dice gain at 512×512 . These results are consistent with the view that small lesions benefit from both explicit prompt conditioning and higher spatial resolution.

One plausible explanation is that small lesions already provide limited visual evidence, and resizing a large radiograph to 256×256 weakens that evidence further. Under this condition, how prompt information is used becomes especially important. In SAM-Med2D, the box prompt primarily acts as a coarse positional cue for mask generation. When the lesion occupies very few pixels after resizing, this cue alone may be insufficient to recover a detailed mask. By contrast, PGA-UNet uses the bounding box as a dense spatial prior that amplifies the lesion region and propagates prompt information through both the encoder and decoder. This design appears better suited to preserving weak local evidence for small lesions. The stronger result of PGA-UNet at 512×512 is also consistent with this interpretation, since higher resolution preserves more lesion detail before prompt-guided decoding.

H. EFFICIENCY ANALYSIS

This experiment evaluates whether the proposed gains are achieved with a practical computational profile. It supports

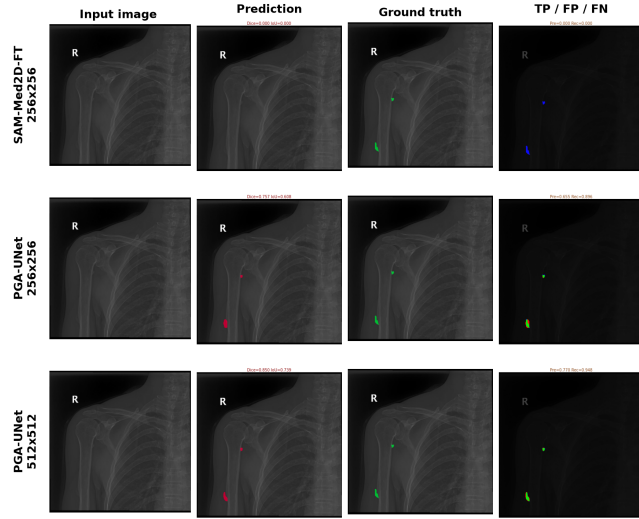


FIGURE 4. Qualitative comparison on small lesions in BTXRD. The examples contrast fine-tuned SAM-Med2D at 256×256 , PGA-UNet at 256×256 , and PGA-UNet at 512×512 .

TABLE 8. Efficiency comparison in the tested environment.

Model	Params	GFLOPs	Size	GPU ms	CPU ms
PGA-UNet 256	2.95M	7.74	11.35 MB	8.17	109.17
PGA-UNet 512	2.95M	30.97	11.35 MB	18.28	336.74
SAM-Med2D 256	271.24M	92.02	2443.16 MB	48.75	1263.77

the claim that the method is suitable for repeated low-latency interactive use rather than only for offline batch inference. Table 8 reports the efficiency comparison, measured on the same NVIDIA Tesla T4 GPU used for training (mean latency over repeated forward passes, batch size 1). PGA-UNet contains only 2.95M parameters. In the tested 256×256 setting, this corresponds to an approximately 92-fold reduction in parameter count and an 11.9-fold reduction in FLOPs relative to SAM-Med2D, together with $6.0\times$ faster GPU inference and $11.6\times$ faster CPU inference.

I. ABLATION STUDY

The ablation study directly tests which proposed components are responsible for the observed gains. In particular, it examines whether performance comes from the prompt representation itself, from encoder-side prompt gating (PSG), from decoder-side prompt-conditioned attention (CAD), or from their combination. We isolate three factors: whether PSG is present in the encoder, what form of decoder skip-attention is used (none, the original unconditioned Attention U-Net gate, or CAD), and whether the prompt is represented as a binary box or a Gaussian-smoothed plateau. Combining these factors gives eight trained configurations per dataset, all at 512×512 : the seven partial variants that isolate individual factors and their pairwise interactions, plus the full proposed PGA-UNet, which uses PSG, CAD, and the Gaussian prompt together. Tables 9 and 10 report Dice under covering and off-center prompts for BTXRD and FracAtlas, respectively.

TABLE 9. Ablation results (Dice) on BTXRD at 512×512 .

PSG	Decoder attention	Prompt	Covering	Off-center
No	None	Gaussian	0.8596	0.8230
No	None	Binary	0.8597	0.8253
No	Vanilla gate	Gaussian	0.8671	0.8408
No	CAD	Gaussian	0.8685	0.8358
Yes	None	Gaussian	0.8567	0.8297
Yes	Vanilla gate	Gaussian	0.8611	0.8314
Yes	CAD	Binary	0.8611	0.8317
Yes	CAD	Gaussian (PGA-UNet)	0.8788	0.8496

TABLE 10. Ablation results (Dice) on FracAtlas at 512×512 .

PSG	Decoder attention	Prompt	Covering	Off-center
No	None	Gaussian	0.7727	0.7356
No	None	Binary	0.7670	0.7199
No	Vanilla gate	Gaussian	0.7560	0.7311
No	CAD	Gaussian	0.8158	0.7777
Yes	None	Gaussian	0.7115	0.6990
Yes	Vanilla gate	Gaussian	0.7372	0.7170
Yes	CAD	Binary	0.8132	0.7896
Yes	CAD	Gaussian (PGA-UNet)	0.8286	0.7925

The full PGA-UNet configuration (PSG, CAD, and the Gaussian prompt combined) is the single best row in both tables, under both covering and off-center prompts, on both datasets. This is the clearest outcome of the ablation study: none of the seven partial configurations matches the full combination in any of the four covering/off-center by dataset settings.

The component-wise comparisons reveal a consistent pattern of synergy rather than three independent additive contributions. First, PSG in isolation, that is, without CAD in the decoder, does not reliably help and can hurt: on BTXRD it leaves Dice essentially unchanged relative to the no-PSG, no-attention baseline (0.8567 versus 0.8596 covering), while on FracAtlas it reduces Dice by 0.0612 covering and 0.0366 off-center relative to the same baseline. Encoder-side prompt amplification without a matching decoder-side mechanism to exploit it can therefore be actively counterproductive. Second, CAD in isolation, that is, without PSG, does help on both datasets (covering Dice rises by 0.0089 on BTXRD and 0.0431 on FracAtlas relative to the no-PSG, no-attention baseline), and it is more reliable than the original unconditioned attention gate: replacing CAD with the vanilla gate in the same no-PSG setting costs 0.0167 covering Dice on FracAtlas, where the unconditioned gate falls even below the no-attention baseline. Third, once PSG and CAD are combined, the pairing outperforms either component alone by a wide margin, most visibly on FracAtlas, where the full model exceeds the CAD-only variant by 0.0128 covering and 0.0148 off-center Dice, and exceeds the PSG-only variant by 0.1171 covering and 0.0935 off-center Dice. This indicates that PSG's benefit is conditional on CAD being present to make productive use of the amplified encoder features, rather than PSG being independently useful.

The prompt-representation comparison shows a related pattern. Replacing the Gaussian prompt with a binary prompt

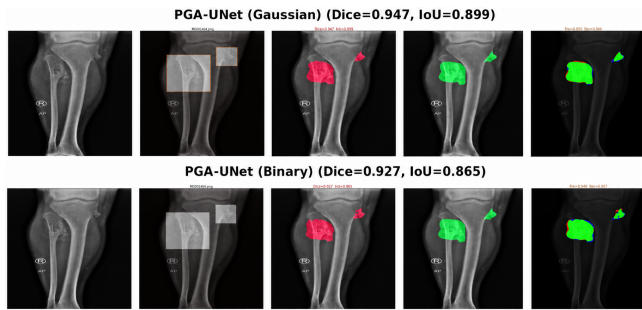


FIGURE 5. Qualitative comparison between the Gaussian prompt representation and the binary prompt representation. The Gaussian version produces cleaner boundaries and better overlap on this representative BTXRD example.



FIGURE 6. Representative failure case in an anatomically overlapping region. Even with a covering prompt, the lesion boundary remains difficult because the local X-ray structures are visually confounded.

inside the full PSG+CAD architecture costs 0.0177 covering and 0.0179 off-center Dice on BTXRD, and 0.0154 covering and 0.0029 off-center Dice on FracAtlas. At the no-PSG, no-attention baseline, however, the same substitution has almost no effect (within 0.006 Dice on BTXRD, and a smaller 0.0057–0.0157 Dice gap on FracAtlas that favors Gaussian but is far below the gap observed in the full architecture). The Gaussian plateau representation therefore becomes clearly beneficial only once the network is designed to reuse prompt information throughout the encoder and decoder, again pointing to synergy between the three design choices rather than to any one of them being independently sufficient.

J. FAILURE MODES

The method still has identifiable failure cases. In the qualitative review collected during the thesis experiments, lower Dice scores were more likely for lesions with highly elongated or branching shapes, lesions located in anatomically cluttered regions such as the shoulder or clavicle, and lesions with weak contrast relative to surrounding structures. These cases are relevant because they show that prompt guidance does not eliminate ambiguity once the local image evidence itself is poor.

VI. DISCUSSION

The results support several main observations. First, PGA-UNet performs well on both datasets, and Monte Carlo cross-validation shows that this performance is stable across repeated random image-level splits rather than tied to one favorable partition. Second, compared with the stronger automatic baseline Attention U-Net, PGA-UNet achieves higher system-level segmentation performance when coarse box guidance is available, and the gap is largest precisely on the

images where the automatic baseline is weakest. Third, compared with fine-tuned SAM-Med2D at the matched 256×256 resolution, PGA-UNet again performs better on both datasets, under both covering and off-center prompts, despite being trained only on covering prompts while SAM-Med2D was trained on both. Fourth, the advantage becomes even clearer on the small-lesion subset, where fine-tuned SAM-Med2D nearly fails to produce a usable mask. Fifth, the ablation results show that PSG and CAD are most effective jointly rather than independently, with the full combination outperforming every partial configuration under both covering and off-center prompts on both datasets. Finally, the efficiency analysis shows that the model remains substantially lighter and faster than SAM-Med2D.

The scope of the analysis was deliberately narrowed. The auxiliary screening pipeline is excluded because it addresses a different question from prompt-guided segmentation itself. Likewise, the retained subgroup analyses are objective constructions tied either to lesion size or to a baseline-defined ranking.

The current study still has further limitations. The model depends on resizing inputs to a fixed resolution, which can dilute fine lesion detail in very small targets. The current formulation does not yet exploit additional information sources beyond the image and box prompt, and the training setup still requires dense mask supervision. The evaluation also remains limited to the current public datasets. Prompts are simulated from annotations rather than collected from radiologists, and BTXRD and FracAtlas are evaluated in separate retraining settings rather than through external-domain transfer. The Monte Carlo cross-validation experiments cover the main resolution and prompt-condition trend at 512×512 , but the ablation study itself is still reported from a single deterministic split rather than from repeated multi-seed training.

An additional limitation is interpretive rather than purely empirical. Several observed gains, especially under prompt perturbation, are consistent with the intended roles of PSG, CAD, and the Gaussian plateau prompt, but this article does not include feature-level probing that would establish a unique internal mechanism for those gains. The ablation study therefore supports the usefulness of the design choices at the system level, not a definitive causal account of representation dynamics inside the network.

VII. CONCLUSION

This article presented PGA-UNet, a lightweight prompt-guided CNN for interactive bone X-ray lesion segmentation. By combining a plateau-style Gaussian prompt map with encoder-side Prompt Spatial Gates and decoder-side Conditional Attention Decoders, the model achieves favorable results on BTXRD and FracAtlas while remaining substantially smaller than SAM-Med2D. The most consistent evidence for the design lies in the ablation study, where the full PSG+CAD+Gaussian combination outperforms every partial configuration under both covering and off-center prompts on both datasets; in the small-lesion comparison,

where fine-tuned SAM-Med2D nearly fails while PGA-UNet remains usable; and in greater stability on Attention U-Net-defined failure cases. Future work should evaluate the method with radiologist-provided prompts, external cohorts, higher-resolution processing without aggressive resizing, and broader clinical interaction settings.

DATA AVAILABILITY

BTXRD and FracAtlas are public datasets cited in this article. The manuscript reports experiments on image-level train/validation/test splits derived from those datasets. Code and split details should be released with the final reproducibility package.

AUTHOR CONTRIBUTIONS

Thong Luc: conceptualization, methodology, software, investigation, formal analysis, visualization, writing of the original draft. Nguyen Huu Binh: software support, data preparation, experiment verification, review and editing. Ly Quoc Ngoc: supervision, validation, review and editing.

ETHICS STATEMENT

This study used public de-identified radiograph datasets and did not involve prospective data collection, patient contact, or intervention.

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OpenAI ChatGPT was used only for vocabulary and grammar refinement in selected non-technical manuscript text. All technical content, experimental design, mathematical formulation, figures, tables, and conclusions were prepared and verified by the authors, who take full responsibility for the final manuscript.

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